CORAL BLEACHING DETECTION USING DEEP LEARNING

1. WHAT IS THE PROJECT?

Coral reefs turn white when they are stressed by hot water. This is called coral bleaching. Checking reefs by hand takes a lot of time.

This project teaches a computer to look at a coral photo and tell whether the coral is bleached or healthy.

In simple words: Input = coral image -> Output = coral condition

2. DATASET

- Name: NOAA PIFSC ESD Coral Bleaching Dataset
- Source: Hugging Face (NMFS-OSI), made by NOAA
- Contains coral photos sorted into class folders
- The folder name is used as the label of each image
- Split: 80% for training, 20% for validation (testing)

3. MODEL USED

ConvNeXt-Tiny (pretrained)

- It is a CNN (Convolutional Neural Network) for images.
- It was already trained on ImageNet (over 1 million images), so it already knows edges, colors and textures.
- I reuse that knowledge and fine-tune it on coral images. This is called TRANSFER LEARNING. It gives good accuracy with less data and less training time.
- Loaded using the "timm" library.

4. HOW IT WORKS (STEP BY STEP)

Step 1: Download the dataset from Hugging Face.
Step 2: Read all images and take labels from folder names.
Step 3: Split into training and validation sets.
Step 4: Prepare images:
    - Resize to 224 x 224
    - Training images get random flips, crops and color changes (data augmentation) so the model does not just memorize.
    - Normalize colors the way ImageNet models expect.
Step 5: Train the model for 10 epochs on a GPU (Google Colab T4).
Step 6: After every epoch, check accuracy on validation images and

save the best model (best_model.pt).
Step 7: Show results: graphs, classification report, confusion matrix.
Step 8: Load the saved model in a new notebook, give an image path,
and get the prediction with a confidence percentage.

5. TRAINING SETTINGS

- Loss function : Cross Entropy (with label smoothing)
- Optimizer : AdamW
- Learning rate : OneCycle scheduler (changes during training)
- Mixed precision (float16) for faster training on GPU
- Batch size : 32
- Epochs : 10

6. RESULT

- Best validation accuracy: about 89.3%
- Training loss went down from 0.45 to 0.26
- Accuracy stopped improving after about epoch 5, so the best model was saved from that point.

7. EXAMPLE OF USE

Give a coral image -> the model prints something like:
Predicted: bleached (92.5%)
It also shows the probability for every class.

8. TOOLS AND LIBRARIES

- Python
- PyTorch and torchvision (deep learning)
- timm (pretrained ConvNeXt model)
- scikit-learn (data split, reports, confusion matrix)
- Matplotlib (graphs and image plots)
- Pillow (image loading)
- Google Colab with T4 GPU
- Hugging Face Hub (dataset download)

9. USES

- Faster monitoring of coral reef health
- Helps marine scientists study the effect of climate change
- Can be used in underwater drones or mobile apps in the future

10. LIMITATIONS AND FUTURE WORK

- Accuracy is around 89%, and can improve with bigger images, a bigger model, or more data.
- Some bleaching levels look very similar, so mistakes happen between close classes.
- Future: build a mobile or web app and use real-time underwater camera images.

ONE-LINE SUMMARY

I fine-tuned a pretrained ConvNeXt-Tiny CNN on NOAA coral images to automatically detect coral bleaching with about 89% accuracy.